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EDUCATIONAL BACKGROUND

Nanyang Technological University (QS12) **Singapore**
Master of Science in Signal Processing and Machine Learning 2026.01-2027.07
Chongqing University of Technology **Chongqing, China**
Bachelor of Science in Computer Science and Technology ***(Honours)*** 2021.09-2025.06

- **GPA:** 3.96 / 5.0 (**Top 1%**)

INTERNSHIP EXPERIENCE

Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm) 2026.05 ~ 2026.08
Research Intern in Embodied AI for Robotics
Project Overview: Successfully adapted and deployed the **state-of-the-art dual-system navigation model DualVLN (based on InternVLA-N1)** onto the **Lenovo Morningstar MX robot**. The **decision-making layer (System 2)** fine-tunes **QwenVL-2.5 (7B)** to generate **2D pixel-level sub-goals**, while the **control layer (System 1)** utilizes a **Flow-Matching (NextDiT)** model to predict **32-step continuous physical trajectories**, thereby establishing a **closed-loop pipeline** spanning semantic parsing, pixel-level sub-goal localization, and high-frequency trajectory execution on the physical robot.

• **Asynchronous Dual-System Inference:** An **asynchronous cascading mechanism** between the decision-making and execution layers extracts **semantic features of sub-goals** from the Large Model's output and projects them into the control layer as **trajectory guidance conditions**. **TensorRT optimization** at the edge reduces **S1 trajectory generation time to 30 ms**, while **KV-Cache state reuse** compresses **S2 latency from 1.1 s to 0.7 s**, eliminating physical stuttering caused by frequent Large Model calls on the physical robot.

• **Depth-Agnostic Physical Resilience:** To address **depth sensor failures** caused by transparent media or direct strong light, an **adaptive depth-agnostic pass-through scheme** was designed. When sensor data is anomalous, the system automatically bypasses the **S1 trajectory flow matching and denoising** steps, directly passing the **2D pixel sub-goals** predicted by **S2** to the planning and control layer. Without training on low-level trajectories, this scheme achieved a **51.60% zero-shot transfer success rate** in physical simulation tests, reducing **localization error to 4.66 m**.

• **Active "Look-Down" Self-Check:** To mitigate **camera field-of-view limitations** and **physical blind spots**, an **active "look-down" self-check mechanism** was introduced. **Downward-tilted images** are injected in real time into the **VLM temporal sliding window** to perform **non-blocking secondary sub-goal planning** and **blind-spot checks**. The system achieved a **64.3% success rate** in standard **VLN-CE R2R** tests, significantly reducing the risk of **blind-spot collisions** in complex physical environments.

PROJECT EXPERIENCE

OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision 2026.01-2026.03
First Author, **IEEE RA-L (SCI Q2 TOP,** Under Review) [arXiv] Supervisor: Prof. Lihua Xie
Project Overview: Addressing the issues of **missing spatial reasoning** and **context explosion** in **large-scale 3D scenes** using **LLM frameworks**, a **perceptually driven embodied intelligence agent framework** was developed based on **Python, PyTorch, and ROS2**. By constructing a **dynamic scene graph** as the agent's **world model** and employing a **multi-resolution spatial attention mechanism**, complex **long-range zero-shot navigation** was achieved with a **93.18% success rate**, while reducing **token costs by up to 69.98%**.

• **Intelligent Agent Architecture:** A **stateful task orchestration architecture** was built based on **LangGraph**, deeply integrating **multimodal large models** with underlying control skills through a **Function Calling** mechanism. Employing the **MCP** approach, environmental knowledge is persistently managed and retrieved on demand through a **3D scene graph**. A **reflective Actor-Critic decision-making loop** was designed, and **hierarchical Chain-of-Thought (CoT) reasoning** enabled **multi-resolution semantic space navigation**, improving the **long-range navigation success rate by 15.91%**.

• **Hierarchical World Model:** A **five-layer dynamic scene graph** was constructed. **Object instantiation** under omnidirectional vision was achieved using **SAM2**, and **adaptive room partitioning** was implemented based on a **continuous cohomology algorithm**, completing the semantic leap from **low-level point cloud geometry** to **high-level abstract spatial representations**. A **geometry–VLM hybrid pruning technique**, combined with **KDTree ray detection**, **online multimodal verification**, and **Open3D**-based **high-frequency point cloud processing**, ensured the physical authenticity of the scene graph's **topological connections**.

• **Multi-resolution Spatial Attention:** An innovative **3D stereo eight-neighbor observation model** centered on the robot was proposed, transforming complex coordinate systems into **human-intuitive spatial descriptions**. An **adaptive resolution prompting mechanism** was designed to generate **fine-grained captions** for focus areas while performing **semantic summarization** for peripheral and far-field regions, reducing **token overhead by nearly 70%** while maintaining **decision accuracy**.

AWARDS & HONORS

National Scholarship (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

TECHNOLOGY STACK

• **Embodied AI & LLM Agents:** Proficient in designing **state-machine networks** using **LangGraph/LangChain**; deep understanding of the **MCP protocol** with experience in encapsulating **tool ecosystems**; familiar with **Actor-Critic** decision frameworks, **Chain-of-Thought (CoT)** reasoning, and techniques for **LLM prompt compression** and **adaptive resolution**.

• **Robotic Systems & 3D Perception:** Proficient in **ROS 2** communication mechanisms and **Action/Service architectures**; familiar with **SAM2** object segmentation, **Open3D** point cloud processing, and **KD-Tree** spatial search algorithms; experienced in constructing **3D dynamic scene graphs** and designing **hybrid geometric-topological routing**.

• **Systems Engineering & Concurrent Programming:** Proficient in **Python** and **C++**; familiar with **Linux**, **Docker** containerization, and **Git** version control; experienced in encapsulating **low-level control logic** for **multi-threaded applications** and conducting **hardware-software integration and debugging**.